

Individualized model discovery: The case of anemia patients[☆]

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ABSTRACT

The universal sequel to chronic kidney condition (CKD) is anemia. Patients of anemia have kidneys that are incapable of performing certain basic functions such as sensing of oxygen levels to secrete erythropoietin when red blood cell counts are low. Under such conditions, external administration of human recombinant erythropoietin (EPO) is administered as alternative to improve conditions of CKD patients by increasing their hemoglobin (Hb) levels to a given therapeutic range.

Presently, EPO dosing strategies extensively depend on packet inserts and on “average” responses to the medication from previous patients. Clearly dosage strategies based on these approaches are, at best, nonoptimal to EPO medication and potentially dangerous to patients that do not adhere to the notion of expected “average” response. In this work, a technique called semi-blind robust identification is provided to uniquely identify models of the individual patients of anemia based on their actual Hb responses and EPO administration. Using the *a priori* information and the measured input-output data of the individual patients, the procedure identifies a unique model consisting of a nominal model and the associated model uncertainty for the patients. By incorporating the effects of unknown system initial conditions, considerably small measurement samples can be used in the modeling process.

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1. Introduction

Anemia is a common complication among patients of chronic renal failures. This condition is associated with several

complications such as insufficient production of erythropoietin, a glycoprotein promoting the growth of red blood cells in the bone marrow. Until the discovery of recombinant human erythropoietin (EPO) over 30 years ago [1], anemia management primarily involved repeated blood transfusions

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– a procedure known to be associated with several health related complications such as infections, allergic reactions [2], and allosensitization that decreases the likelihood of a successful kidney transplant. Exogenous administration of EPO has been shown to slow the progress of anemia of end stage of renal disease (ESRD) [3–5]. However, chronic dosing of EPO is a challenge due to large patient variability in erythropoietic responses. Consequently, decision processes involved in EPO dosage strategies consist mostly of trial-and-error [6]. Therefore, a procedure capable of providing clear and consistent dosage-response method for individual patient based uniquely on Hb and EPO responses could provide better dosing strategy for the individual patients. We propose patient specific modeling technique rooted in feedback control theory to provide models based on individual patient information.

Several attempts to automate EPO delivery have already been reported in the literature [7–20]. Bayesian network drug delivery optimization was performed with patient population data in [8]. This approach was subsequently enhanced by the Fuzzy rule-based control strategies in [15]. In [16,20], Artificial Neural Network modeling techniques were evaluated in anemia patient model developments. Both of these modeling techniques rely on general patient population data. Thus none of the above procedures are able to provide models based on the individual patient measurement information. Few attempts have been made towards the development of individualized patient models [9,19,18,21]. Support vector regression [21] and approximate dynamic programming in [9,19,18] have been proposed previously to personalize EPO dosing. The modeling technique explored in [21] mainly focused on predicting input EPO dosage rather than output Hb measurements. The objective of developing model to inputs, as opposed to outputs, in clinical settings is less desirable since predicting the latter provides a means of determining the former with a properly developed model. In [9,19,18] individualized anemia management was attempted; however, the approach was aimed at optimizing the EPO dosage rather than developing individualized patient models.

Many therapeutic solutions are closely related to system control and feedback measurement problems [22–24]. In the case of drug dosing therapy, physicians wish to achieve and stabilize the responses of certain medications over some desired therapeutic range while minimizing control efforts either to minimize costs or avoid toxic side effects related to large use of the medication. From the views of control engineer, this objective can be translated into a feedback control problem. Basic requirements to translate drug dosing problems to the feedback control problem are the following three components: the model of the plant (patient), control input (EPO dose), and output measurements (hemoglobin concentration). In this context, the model defines the mathematical relationship between the input and measurement output values. Clearly, the only missing piece to successfully translate the problem from dosage therapy domain to the feedback domain is the model of the system. The focus of this work is to provide such missing piece by providing a modeling technique to identify such a mathematical relation linking patients' output hemoglobin to input erythropoietin consequently allowing the implementation of feedback control design techniques.

The technique provided in this work differs from the classical approach to system modeling, namely, *system identification* [33]. A major difference between the two modeling techniques is that in the classical method, it is required that a model structure, with unknown parameters, of the system of interest to be predefined. Here, the aim is to estimate these unknown parameters by using the measurement data while assuming that the predefined structure accurately defines the dynamic profiles of the true system. Furthermore, it is assumed that any discrepancy between the behavior of the model and that of the true system is attributed to noise in the data, which is further assumed to have known statistical properties such as the mean, variance, etc. Clearly, the most challenging and time consuming stage in this approach is the selection of the model structure. Selecting a structure with orders higher than the true system's model order usually result in "over-fitting". On the contrary, selecting model order lower than the true system order results in "under-fitting". Therefore, when the selection of the model structure and/or the statistical assumptions on the system becomes questionable, a modeling technique with no *a priori* model structural requirements on the system is a sound alternative [25]. This modeling technique makes no assumptions on neither the structure of the system nor the noise affecting measurements. It only assumes the system belongs to a certain model class (details on this is in the later Section 3) and the noise affecting measurements is assumed unknown but bounded by some value. This bounded noise value is usually available and provided by measuring instrument manufacturers. The result is a model set consisting of models able to explain the *a posteriori* measurement data under given *a priori* assumptions. The center of this model set can now be considered as the nominal model. We previously introduced robust identification procedure to individualized anemia patient model discovery in [26]. This approach did not account for the fact that the system (patient) was already in progress prior to identification and therefore the initial conditions are not zero. Ignoring such information may artificially inflate the model prediction error. Therefore, the aim of this paper is to incorporate such information into the modeling process, consequently providing models with high predictive performance. We refer to this approach as *semi-blind* robust identification [27].

Organization of the paper is as follows. In Section 2, we provide an overview of anemia management problems while pointing out a few of its challenges especially regarding individualized model discovery. Section 3 provides a brief overview on robust identification, especially in ℓ_1 identification framework, subsequently introducing *semi-blind* robust identification procedures in the same section. We combine the robust identification techniques developed in Section 3 with anemia management problems in Section 2 to produce nominal individualized model in Section 4. Section 5 concludes the paper.

2. Anemia management problem

2.1. Introduction

In a healthy human, about 10^{10} red blood cells (RBC) are synthesized per hour in order to maintain sufficient hemoglobin

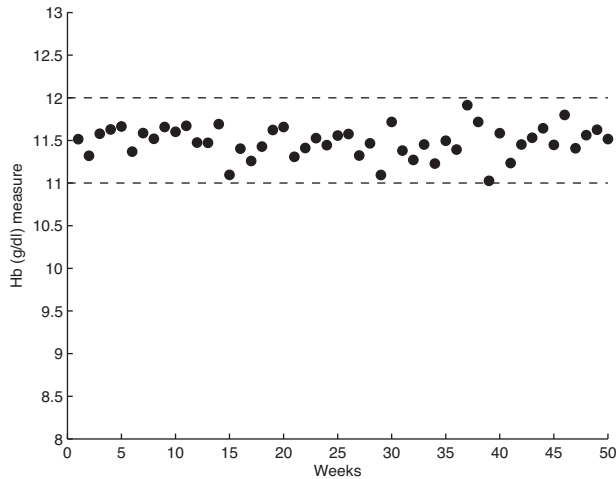


Fig. 1 – Example of desirable Hb concentration during EPO treatment.

levels of 13–14 g/dL [28]. This process is driven by erythropoietin, a hormone produced primarily in the kidney cortex [3]. In chronic kidney disease (CKD) patients, natural production of this hormone is compromised leading to chronic anemia. The current treatment of choice for patients with such condition is external administration of human recombinant erythropoietin (EPO). The main challenge is the amount of EPO to administer to a given patient. The existing dosing strategies, more or less, involve the responses of the “average” patient to the medication. However, inter-patient and intra-individual variability in erythropoietic responses makes EPO dosing based on such strategy difficult and potentially dangerous.

2.2. Definition

The primary goal of dosing EPO is to achieve and stabilize hemoglobin concentration of ESRD patients at a level that minimizes the need for blood transfusions. The National Kidney Foundation’s Dialysis Outcomes Quality Initiative defined that range to be between 11 and 12 g/dL [29].

To achieve this objective, several dialysis facilities have developed their own protocols for EPO dosage adjustments [30]. These protocols generally combine dosing information included on FDA-approved EPO packet inserts with physician’s own experience to determine dosing amounts. Such protocols tend to address the dosing as a population-wide issue. However, the inherent variability between patient heterogeneity in the response to the drug renders such an approach less than effective. Fig. 1 shows an example of the desired outcome of anemia treatment with EPO. Here, the two dotted horizontal lines show the upper and lower bounds on the desired hemoglobin responses to EPO dosing. The dotted dark circles show the hemoglobin values within the desired range of 11–12 g/dL. Fig. 2 shows a typically observed hemoglobin outcome. Clearly, the Hb responses shown in this figure missed the desired target range considerably. A possible explanation behind this behavior is a phenomenon known as hemoglobin cycling which results when patients’ hemoglobin measures

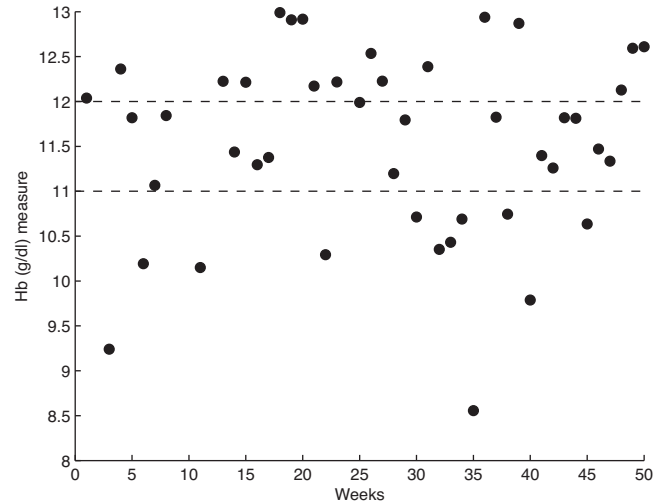


Fig. 2 – Example of undesirable Hb concentration during EPO treatment.

overshoots and undershoots the targeted range due to EPO dosing strategies based on “expected” outcomes. Over 90% of ESRD patients suffer from this phenomenon during EPO therapy [31].

To address this challenge, patient-specific approach to dosing strategies is necessary; then, modern feedback control techniques can be used to develop patient patient-specific treatment strategies.

2.3. Control problem

Fig. 3 shows a typical configuration of the interaction between a physician and an ESRD patient in a feedback loop. In this figure, Hb_d represents the desired hemoglobin level in a steady state. The amount of medication, EPO, to administer is determined by the feedback controller. Fluid volume changes, undiagnosed blood loss, and acute conditions can be the source of *disturbance* in the system. In addition, since the actual hemoglobin Hb values cannot be accurately measured, the measured hemoglobin output values Hb_m are assumed to be contaminated with *noise*, as shown in the figure. The *error* term reflects the discrepancies between the measured hemoglobin response and the desired values. Using feedback control strategies in drug optimization therapy has long been

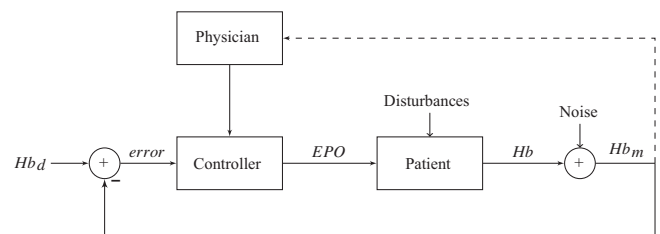


Fig. 3 – Closed-loop feedback system representing patient and physician interaction with erythropoietin and hemoglobin where Hb_d is the desired hemoglobin level and Hb_m is the measured hemoglobin level.

recognized in the literature [22]. Basic requirements for such an approach include:

- (i) a clear design objective (stabilize hemoglobin level within the target range, for example 11–12 g/dL)
- (ii) controller (algorithm determining the optimal input EPO to the patient)
- (iii) plant (patient)
- (iv) plant model – a mathematical equation relating input-output (EPO to Hb)
- (v) output (Hb response) measurements

Clearly, the most important piece of information is a patient specific mathematical model of the dose-response relationship between EPO and Hb. Developing a methodology capable of producing individualized patient models will complete the needed requirements to fully translate AMP to feedback control problem.

A major challenge to modeling in clinical environment is the lack of measurement samples. This data insufficiency results primarily since patients often neglect to follow up on appointments and/or switch medications. Therefore, a modeling technique which only considers small sample data could be beneficial. Modeling techniques using the classical identification methods could not develop models based on small measurement samples due to the statistical assumptions in this modeling techniques [32].

The robust identification approach employed here makes no assumption on the nature of noise affecting measurements nor on the order of the system. These properties make it capable of developing individualized models from sparse data (as demonstrated in Section 4). It should be mentioned that model-free based control algorithms are available and can be used [9] to achieve similar objectives of stabilizing hemoglobin levels. However, such approach lacks an important benefit of providing a clear mathematical relations linking EPO to Hb response as provided in model based approaches.

3. Robust identification

3.1. Introduction

We provide a brief introduction to robust identification. First, we provide an overview on system modeling based on measurement samples using the classical identification techniques, then we follow that with the robust identification method.

3.2. System identification

The conventional system identification approach to model discovery such as the Autoregressive with Exogenous (ARX), Prediction Error Estimate (PEM), Output Error (OE), etc. (see [33] for details) involve

- (i) assumption on fixed mathematical model structure with known order and unknown parameters
- (ii) definition of performance measure

- (iii) use of measured data to estimate the unknown parameters
- (iv) validation of the model with unseen data.

Usually, the least squares techniques are used to determine these unknown parameters. However, least squares methods are known to be sensitive to outliers, hence the use of such procedure is intrinsically non-robust. An issue with this approach is that it assumes the error affecting measurements is known to be of certain statistical characteristics, usually of white Gaussian noise nature. In addition, the classical identification technique assumes a fixed mathematical model structure that can explain the system dynamics regardless of its complexity. This technique is suitable for many physically-driven systems; however, when applied on complex system such as human physiological systems, the technique becomes a challenge. It is well known in the system identification communities that one of the most difficult, and time consuming, stages in system modeling from measurement sample is the selection of model structure. Selecting model with orders higher than that of the true plant's order results in "over-fitting" the data whereas selecting model class with orders less than that of the true system order results in "under-fitting". When the selection of model structure is unrealistic and/or the error affecting measurements is questionable, robust deterministic identification approach is a sound alternative [25].

3.3. Robust identification

The robust identification technique accounts for the fact that information available on the system of interest is finite, partial, and corrupt hence restraining its model to a fixed structure may be erroneous or short lived. The technique relies on both functional analysis and interpolatory theory as well as the theory and concepts of information-based complexity (IBC) in which a solution to a continuous system is sort after using discrete measurement information (see [34] and references therein). Information is *partial* because there is no complete information available on the problem at hand, *finite* since the only available measurement data is finite, and *corrupt* to reflect the fact that the available measurement is corrupted in a way (e.g. by noise). This was initially realized and utilized by Helmicki [35] followed by many others [36–38] when the conventional approaches to model discovery using input-output measurements failed to provide the *hard* bound requirement necessary for modern robust controller synthesis.

An attractive feature of the robust identification method is the minimal *a priori* requirements on the system. There is no assumption on the model order nor statistical assumptions on the noise affecting output measurements. The only assumption is that the system belongs to a particular class and its measurement noise is *unknown but bounded* by a known value, usually provided by measuring device manufacturer. The aim is to obtain a suitable nominal model concurrent with a *hard* uncertainty bound using the *a posteriori* information (experimental input-output data) and the given *a priori* information. Results of the proposed approach can be visualized in Fig. 4, where the center of the circle represents the nominal model while its radius represents the uncertainty in the model. The

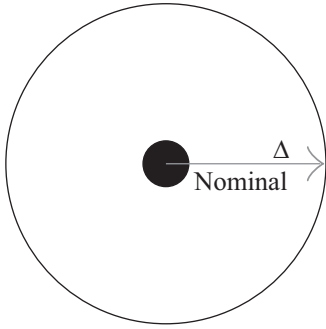


Fig. 4 – Pictorial view of nominal model (center dot) with bounded uncertainty (radius Δ).

derived nominal model can be readily used in any advanced controller design synthesis such as adaptive control, predictive control, intelligence control, and more importantly, robust control which requires a nominal model as well as *hard* bound on model uncertainty. It should be emphasized that the uncertainty bound obtained is a culminated uncertainty due to measurement noise as well as uncertainty in the model itself.

To continue the discussion on the subject, let us establish some few needed notations. Of interest are the subset of functions $\mathcal{H}_\infty$ in $\mathcal{L}_\infty$ which are analytically continuous inside the unit disk equipped with the norm $|G|_\infty \doteq \text{esssup}_{|z|<1} \tilde{\sigma}(G(z))$ where

$\mathcal{L}_\infty$ are Lebesgue functions and $\tilde{\sigma}(G(z))$ is the maximum singular value of system $G(z)$. For $G(z) \in \mathcal{H}_\infty$, define $G(z) \in \mathcal{BRH}_\infty$ as bounded real rational transfer functions in $\mathcal{H}_\infty$. Also of interest is space of transfer matrices analytic in the $|z| \leq \rho$ equipped with the norm $|G|_{\infty,\rho} \doteq \text{esssup}_{|z|<\rho} \tilde{\sigma}(G(z))$ defined as $\mathcal{H}_{\infty,\rho}$.

The right-sided Z-transform of real sequence $\{x\}$ is defined as $\mathcal{X}(z) = \sum_{i=0}^{\infty} x_i z^i$.

Given an input-output data sequence, we define an operator G as a function mapping the input sequence to the output. This representation can be defined either as a (rational) complex valued function:

$$G(z) \doteq \sum_{i=0}^{\infty} g_i z^i$$

or as a minimal state-space realization:

$$G \equiv \begin{pmatrix} A & B \\ C & D \end{pmatrix}$$

In sequel, for stable system G , we denote $T_G: \ell^\infty[0, \infty) \rightarrow \ell^\infty[0, \infty)$ the Toeplitz matrix and $\Gamma_G: \ell^\infty(-\infty, -1] \rightarrow \ell^\infty[0, \infty)$ the Hankel operator associated with G .

The following result is used to establish the existence of a linear-time invariant (LTI) systems subject to the given *a priori* information.

Theorem 3.1. (Carathéodory-Fejér). Given a matrix valued sequence $\{L_i\}_{i=0}^{N-1}$, there exists a causal, discrete-time, LTI operator $L(z) \in \mathcal{BH}_\infty$ such that

$$L(z) = L_0 + L_1 z + L_2 z^2 + \dots + L_{N-1} z^{N-1} + \dots \quad (1)$$

if and only if $(T_L^N)^T T_L^N \leq I$ [39]

where $L(z) \in \mathcal{BH}_\infty$ defines bounded functions analytic in the unit circle while (T_L^N) and $(T_L^N)^T$ are, respectively, the Toeplitz and its conjugate transpose and I is the identity matrix with the appropriate dimensions.

3.4. The model

The aim is to derive a discrete linear-time invariant (LTI) model relating hemoglobin concentration to erythropoietin administration using the available *a priori* and *a posteriori* information. The available *a priori* information on the model consists of three positive constants: $K > 0$, $\rho > 1$, and $\epsilon > 0$, which respectively are, the upper bound on a certain gain associated with plant, the lower bound on the relative stability margin of the plant, and the upper bound on measurement noise. From the time domain measurement data, the Carathéodory-Fejér interpolation theory of Theorem 3.1 is used to establish such models. The aim is to determine the necessary and sufficient conditions for there to exist a function $h \in \mathcal{BRH}_\infty$ such that

$$h(z) = h_0 + h_1 z + h_2 z^2 + \dots + h_n z^n + z^{n+1} h^n(z) \quad (2)$$

for some $h^n(z) \in \mathcal{BRH}_\infty$, where $h \in \mathcal{BRH}_\infty$ belongs to the class of real valued analytic functions bounded in the unit circle. Then from [39], such a function, h , exists if and only if:

$$I - HH^* \geq 0 \quad (3)$$

where H is the Toeplitz matrix defined in terms of impulse response h , H^* is its conjugate transpose, and I is the identity matrix with proper dimensions [40].

With the *a priori* assumptions, it is assumed that the system generating the *a posteriori* experimental data belongs to the following system class of models, $\mathcal{S}$, such that

$$\mathcal{S} \doteq \{G(z) = H(z) + P(z)\} \quad (4)$$

where $H(z) \in \mathcal{BH}_{\infty,\rho}(K)$ for some $\rho > 1$ representing the non-parametric and $P(z)$ the parametric portion of the system, respectively. Furthermore, we assume the parametric portion of the system, $P(z)$, belongs to some affine set $\mathcal{P}$ of the form:

$$\mathcal{P} \doteq \{P(z) = p^T G_p(z), p \in \mathcal{R}^{N_p}\} \quad (5)$$

where $p \in \mathcal{R}^{N_p}$ are some unknown vectors of some *a priori* known component G_p and N_p is the number of unknown p parameters to be determined.

The following lemma is used to establish the existence of a family of systems in $\mathcal{S}$.

Lemma 3.2. Given K , ρ , and two other vector sequences (u, y) representing input and output measurements, respectively, there exists an operator $S \in \mathcal{S}$ such that $y = Su$ if and only if there exists a vector h such that the following conditions are satisfied [41]:

$$M(h) = \begin{bmatrix} KR^{-2} & (T_h^N)^T \\ T_h^N & KR^2 \end{bmatrix} \geq 0 \quad (6)$$

$$|y - (T_u^N p P + T_u^N h)| \leq \epsilon$$

where $(P)_k \triangleq [g_k^1 g_k^2 \dots g_k^{N_p}]$, g_k^i denotes the k th Markov parameter of the i th transfer function of the parametric portion and h_k are the Markov parameters of the nonparametric portion. T_u^N defines the Toeplitz matrix of N inputs u , T_h^N defines the Toeplitz matrix corresponding to the non-parametric portion of the system response h , while

$$R = \text{diag}[1 \rho \rho^2 \dots \rho^{N-1}]$$

is the diagonal matrix formed by the stability margin, ρ . Since the model is assumed to be in the a priori model class of S which contains several other models capable of generating the measurement data with the assumed bounded noise, we can determine the nominal model by parameterizing the set S with a free parameter $Q(z) \in \mathcal{BH}_\infty$. A simple such parameter of choice is the zero parameter, thus $Q(z)=0$ leads to the central model $S_{\text{central}} = H_0(z) + p^T G_p(z)$ where explicit state space realization of $H_0(z)$ can be determined [42].

3.5. Semi-blind robust identification

For individualized ESRD erythropoiesis therapy, our hypothesis is that by accounting for the effects of patients initial conditions, we can significantly enhance the predictive ability of the derived models. In the robust identification process introduced in Section 3.3, we identify the system G by assuming that the experimental data are generated exactly from some initial time t_0 while ignoring the effects of basal erythropoiesis inputs prior to t_0 . Neglecting such information could lead to artificially high prediction error as seen in our previous work [26]. To account for the effects of initial conditions, we consider a recently developed semi-blind robust identification [27]. For the purpose of this study, we encapsulate the effect of prior inputs (erythropoiesis) on the system into some unknown non-zero variable x_0 . Subsequently by evoking Lemma 3.2, Eq. (6) can be rewritten as shown in Eq. (7). Here, the K_G parameter is the initial conditions system gain and K_{u^-} is the maximum initial input and are interpreted in a component-wise [27]. The rest of descriptions in the equation are the same as those found in (6).

$$\begin{aligned} M(h) &= \begin{bmatrix} KR^{-2} & (T_h^N)^T \\ T_h^N & KR^2 \end{bmatrix} \geq 0 \\ |y - (T_u^N p P + T_h^N h + x_0)| &\leq \epsilon \\ \|x_0\|_q &\leq K_G K_{u^-} \end{aligned} \quad (7)$$

4. Application and results

In the literature, ESRD patients are classified either as a *normal responder* or *poor responders* to EPO therapy [9,15,43]. Such classification provides the means to estimate the expected hemoglobin responses to medication within the group. Normal responder patients typically respond very well to average dosage medications, which is defined as the expected EPO to administer in six months [9]. Predicting the hemoglobin concentration of this patient group is lesser of a challenge than those patients that belong to the *poor responder* group. We obtained clinical data from the University of Louisville Chronic Dialysis Unit. These patient data contain weekly

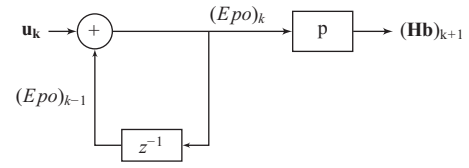


Fig. 5 – Representation of an integrator where u_k is the input to the system. It is the change in EPO level while $(Hb)_{k+1}$ is the output hemoglobin as effected by change in input EPO input.

information of hemoglobin concentration as well as trice weekly EPO dosages. For the identification purposes, the analysis was performed based on weekly data. This implies, the trices weekly EPO dosages are summed to make an effective weekly EPO dosage. By inspecting each patient's hemoglobin profiles in response to EPO dosage, patients are classified as either to belong to the *responder* or the *non-responder* groups. To utilize the semi-blind identification framework proposed in Section 3.5, we need to characterize the possible past input values into a set, $\mathcal{U}_-$. Such set contains all possible EPO (erythropoiesis) values prior to the identification process. In principle, we could assume such a set contains values less than or equal to the maximum possible EPO value ($u^- \in \mathcal{U}_-$, where $u^- \leq |u|_{\max}$); however, such assumption is too coerced and too conservative since the range of EPO dosage varies widely depending on the period of therapy. We can obtain a much tighter bound by considering the effects of *change* in the EPO to hemoglobin levels; consequently the problem becomes identifying an operator S mapping the *change* in erythropoietin, $u_k = \text{Epo}_k - \text{Epo}_{k-1}$ to hemoglobin Hb_{k+1} . With the above assumption, it should be realized that the *a priori* information on the system should therefore include an integrator in the system, subsequently contributing to the parametric portion of the system. Fig. 5 shows such an integrator contribution to the system where u_k is the difference in the current and previous EPO dosage and Hb_{k+1} is the effect of such difference on the patient and p is the constant unknown parameter of the parametric portion of the model $P(z)$. To improve the optimization process of the identification procedure, the input data were uniformly rescaled to the interval $[-1,1]$.

The following *a priori* information on both the system and the noise are used in the study:

$$S = \{G(z) = H(z) + P(z), \quad P(z) = p \frac{z}{z-1}, \quad H \in \mathcal{BH}_{\infty,\rho}(K)\}$$

$$\mathcal{N} = \{\eta \in \ell^\infty : |\eta| \leq \epsilon\}, \quad \epsilon = 0.31 \quad (8)$$

$$\mathcal{U}_- = \{u \in \ell^\infty : |u_k| \leq u_{\max}\}$$

where $P(z)$ and $H(z)$ are the parametric and nonparametric contributions to the system, respectively. The bound on measurement error of $\epsilon = 0.31$ is determined based on informations obtained on hematocrit measuring device used at the dialysis facility [44]. Table 1 summarizes the remaining *a priori* information on the selected patients. The gain, K , is determined based on the LMIs of Eq. (7), the decay rates, ρ , are heuristically determined to obtain reasonable results. Though there may exist some optimization algorithm to determine the

Table 1 – Some a priori parameter information for the selected patients of interesting, patients #1, #28, and #30 and normal responder, patients #6 and #37. N_t represents the number of data points used in the identification and we indicated with a check mark whether patient belongs to a responder or non-responder groups.

Patient#	K	ρ	N_t	Respondent group
1	0.295	1.0604	15	×
28	0.280	1.0900	17	×
30	0.266	1.0700	18	×
6	0.280	1.0214	15	✓
37	0.176	1.0870	15	✓

optimal decay rate values, ρ , such procedure is not known to us at the time of this study.

Fifty-six ESRD patient data were obtained from the University of Louisville Chronic Dialysis Unit for analysis. Typically, each patient data consists of 52 weekly hemoglobin measurements with 0–3 times per week of EPO administration. However, six of the patients have data less than 15 either due to missing appointments or receiving kidney transplants and they were subsequently omitted in this study. This is a common occurrence in medical studies. Using the aforementioned robust identification in conjunction with the semi-blind robust identification procedures, we obtained models ranging from first to third order models for the remaining 50 patients. Due to space limitation, we show the results for five patients: three of which we considered to belong to the non-responder group (Figs. 6–8) and two responder group (Figs. 9 and 10). A one-step-ahead prediction is performed in predicting the hemoglobin levels in each case.

Running the proposed algorithm for the weekly measured hemoglobin data with their corresponding erythropoietin and after performing model reduction by eliminating

uncontrollable and unobservable states, we obtained the following transfer function equations for the selected patients.

$$G^1(z) = \frac{1.03z^3 + 1.398z^2 + 0.8412z - 0.04125}{z^3 + 0.3172z^2 - 0.501z - 0.8161} \quad (9)$$

$$G^{28}(z) = \frac{0.5161z^2 - 0.3610z + 0.472}{z^2 - 0.368z + 0.632} \quad (10)$$

$$G^{30}(z) = \frac{1.018z + 0.2443}{z - 1} \quad (11)$$

$$G^6(z) = \frac{0.8243z^2 + 0.8323z + 0.1615}{z^2 - 0.1139z - 0.8861} \quad (12)$$

$$G^{37}(z) = \frac{1.046z + 0.08025}{z - 1} \quad (13)$$

Using the time delay properties of the z-transform $\mathcal{Z}^{-1}\{z^n G(z)\} = g_{k+n}$, Eqs.(9)–(13) are equivalently represented as follows:

$$\begin{aligned} Hb_{k+3}^1 = & -0.3172Hb_{k+2} + 0.501Hb_{k+1} + 0.8161Hb_k + 1.03u_{k+3} \\ & + 1.398u_{k+2} + 0.8412u_{k+1} - 0.04125u_k \end{aligned} \quad (14)$$

$$\begin{aligned} Hb_{k+2}^{28} = & 0.368Hb_{k+1} - 0.632Hb_k + 0.5161u_{k+2} \\ & - 0.3610u_{k+1} + 0.472u_k \end{aligned} \quad (15)$$

$$Hb_{k+1}^{30} = Hb_k + 1.018u_{k+1} + 0.2443u_k \quad (16)$$

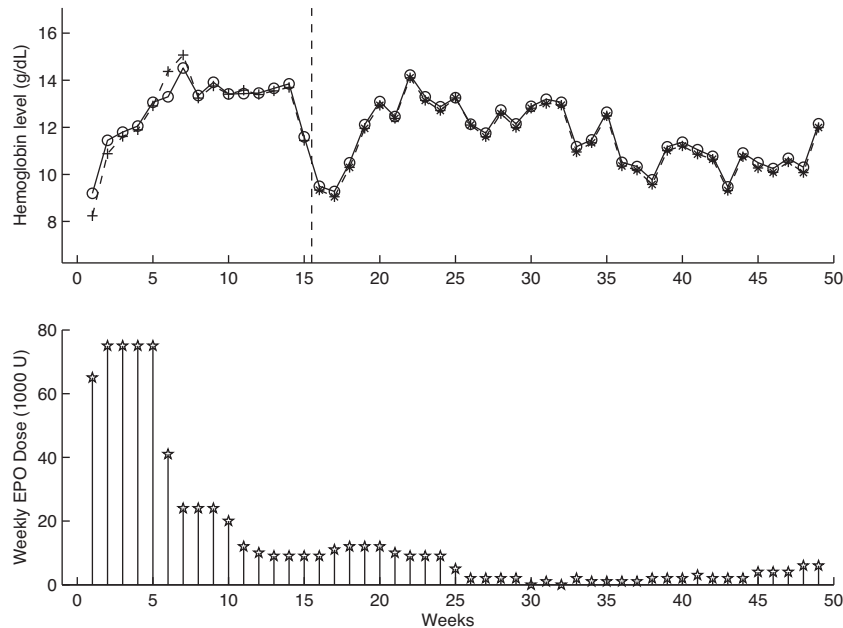


Fig. 6 – Patient #1 [Top] Predicted results with vertical demarcated line separating identification and prediction data. (+) denotes experimental data used in the identification while, (*) denotes additional experimental data not used in the identification. (o) indicates predicted output with 3rd order model. [Bottom] Administered erythropoietin dose in $1000 \times U$ per week.

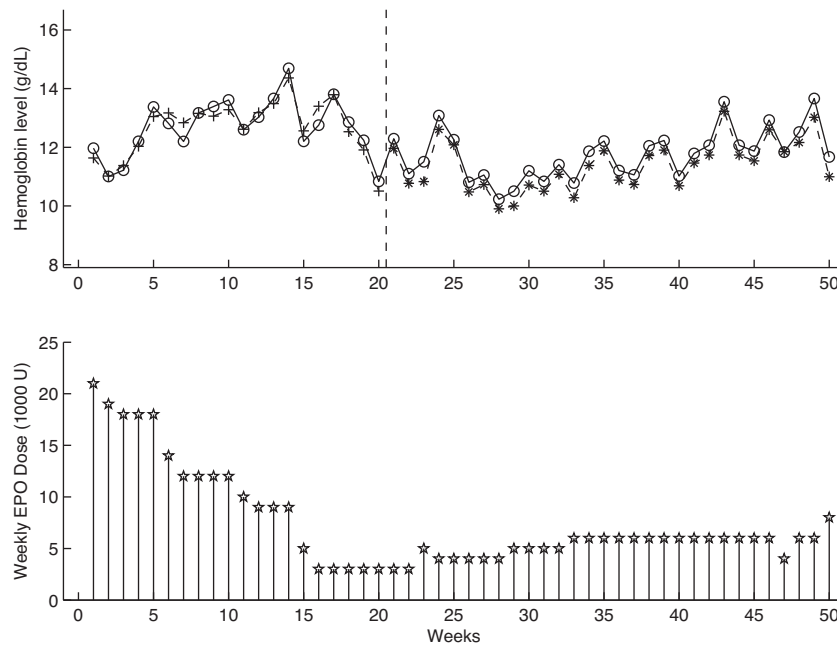


Fig. 7 – Patient #28 [Top] Predicted results with vertical demarcated line separating identification and prediction data. (+) denotes experimental data used in the identification while, (*) denotes additional experimental data not used in the identification. (o) indicates predicted output with 2nd order model. [Bottom] Administered erythropoietin dose in $1000 \times U$ per week.

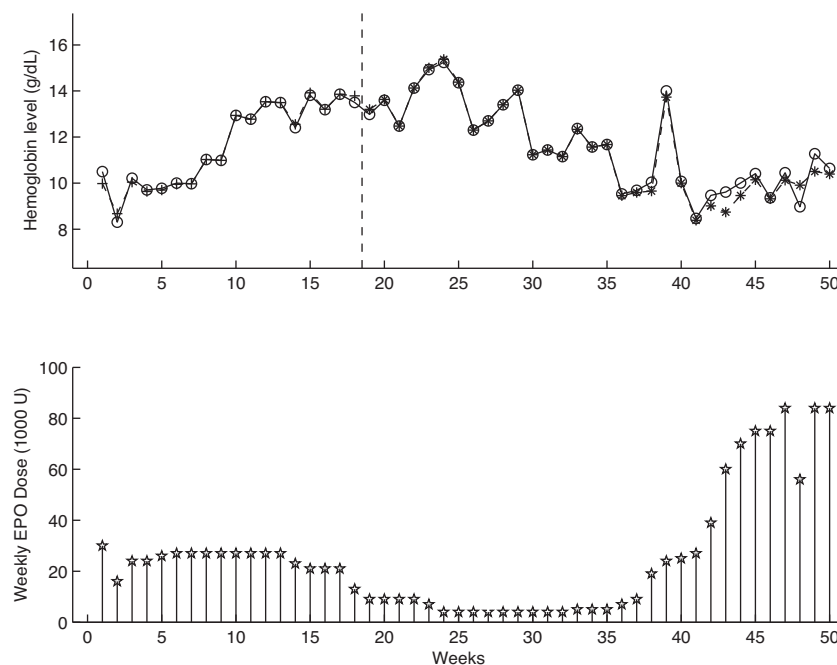


Fig. 8 – Patient #30 [Top] Predicted results with vertical demarcated line separating identification and prediction data. (+) denotes experimental data used in the identification while, (*) denotes additional experimental data not used in the identification. (o) indicates predicted output with 1st order model. [Bottom] Administered erythropoietin dose in $1000 \times U$ per week.

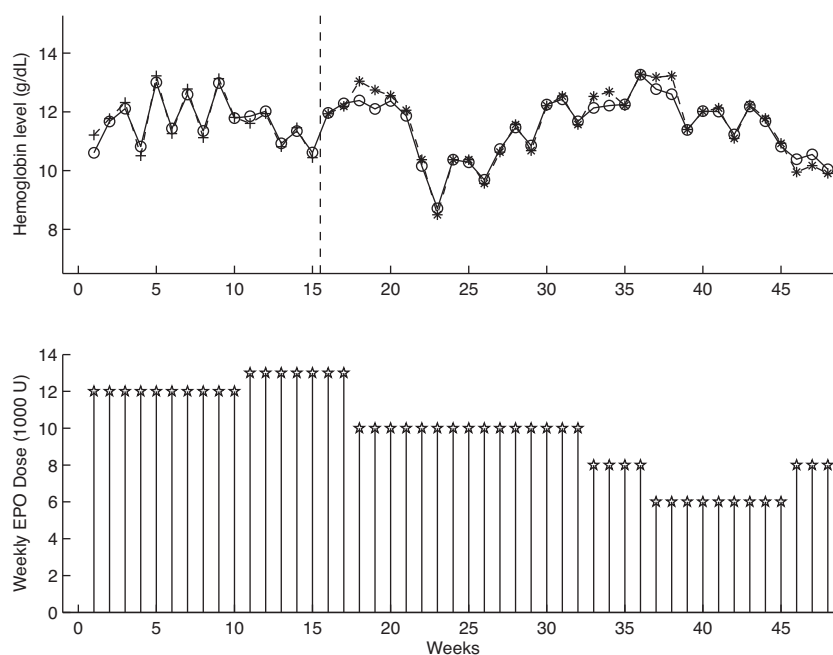


Fig. 9 – Patient #6 [Top] Predicted results with vertical demarcated line separating identification and prediction data. (+) denotes experimental data used in the identification while, (*) denotes additional experimental data not used in the identification. (o) indicates predicted output with 2nd order model. [Bottom] Administered erythropoietin dose in $1000 \times \text{U}$ per week.

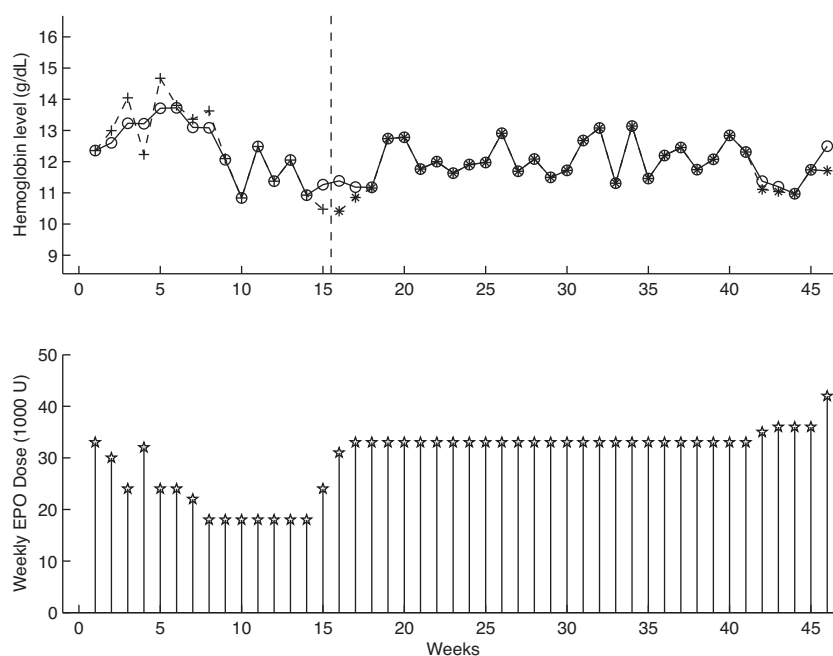


Fig. 10 – Patient #37 [Top] Predicted results with vertical demarcated line separating identification and prediction data. (+) denotes experimental data used in the identification while, (*) denotes additional experimental data not used in the identification. (o) indicates predicted output with 1st order model. [Bottom] Administered erythropoietin dose in $1000 \times \text{U}$ per week.

Table 2 – Model maximum and root mean square error for the selected patients.

Patient#	Error _{max}	Error _{rms}
1	1.07	0.28
28	0.68	0.37
30	1.35	0.40
6	0.65	0.26
37	0.99	0.35

$$Hb_{k+2}^6 = 0.1139Hb_{k+1} + 0.8861Hb_k + 0.8243u_{k+2} + 0.8323u_{k+1} + 0.1615u_k \quad (17)$$

$$Hb_{k+1}^{37} = Hb_k + 1.046u_{k+1} + 0.08025u_k \quad (18)$$

where Hb_k and u_k are the hemoglobin and the change in EPO at time k , respectively.

4.1. Performance measure

The forecasting power of the proposed approach can be seen in Figs. 6–10 for the selected patients of #1, #28, #30, #6, and #37. Fig. 6 shows the predicted hemoglobin responses for patient #1 which corresponds to the discrete Eq. (14). In this figure, “+” denotes the experimental data used for the identification while “*” denotes additional experimental data used for validation purposes. The predicted output of the model is denoted by “o”. In addition, we use a dotted vertical line to demarcate data used during the identification and those used for validation. As shown there, the model is able to predict very well considering only 15 data points are used to generate the model. Similarly, Fig. 8 shows the response of patient #30 using Eq. (15) with 18 input-output data points used for determining the model. Fig. 9 shows similar results with 15 data points used for creating the model for patient #6. The number of data points used for the identification is selected heuristically for optimal predicted results; however, since the procedure is interpolatory, increasing the number of data points used in the identification improves on the predicted results. For each of the selected patients, the maximum prediction error defined in Eq. (19) as well as the average error defined in Eq. (20) are determined,

$$|Error|_{\max} = |y_{data}(i) - y_{pred.}(i)|_{\infty} \quad (19)$$

$$Error_{RMS} = \sqrt{\frac{\sum_{i=1}^N (y_{data}(i) - y_{pred.}(i))^2}{N}} \quad (20)$$

where y_{data} and $y_{pred.}$ are the actual hemoglobin data and model predicted values, respectively. Table 2 provides a summary on both the maximum prediction errors and average errors as well as the number of data points used for the model identification for each of the selected patients. While the maximum prediction error values may seem high, especially for the interesting patients, it should be emphasized that there were no patient specific attributes (weights, race, gender, etc.) considered when determining these models. From the root means square prediction error, however, one can conclude

those error values are acceptable given the width of the target range (11–12 g/dL).

5. Conclusion

This paper addressed issues of developing patient specific models based on their unique measurement values. Specifically, a modeling procedure is provided to develop patient specific models for patients with chronic kidney condition based on their unique hemoglobin response to erythropoietin medication. Using robust identification combined with the recently developed *semi-blind* robust identification techniques, the procedure provides a dynamical model capable of predicting individualized response to EPO medication. A benefit of the methodology is that it requires a very minimal dataset and *a priori* information in developing the models, which makes it useful in clinical settings where, often times, very little information is known about a given patient. It is shown that the technique can provide models based on small sample size as shown in Figs. 6–10, which is an endearing attribute in modeling clinical data. Results for each patient model showed an acceptable predicting error, even though other patient attributes such as weight, demography, race, etc. were not considered in the study. The obtained result is important because it suggests that the robust identification techniques can be used in clinical environments to enhance personalized drug dosage optimization efforts in dialysis units.

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